

SigmaStar TOOL INTRODUCTION

Version 1.0

1. Sstar System Tool

Software developers must use Debug tools and Sstar System tools to access SigmaStar chip registers. Debug tools is shown below.

[Click here for the purchase link](https://item.taobao.com/item.htm?id=15437784110) [https://item.taobao.com/item.htm?id=15437784110]

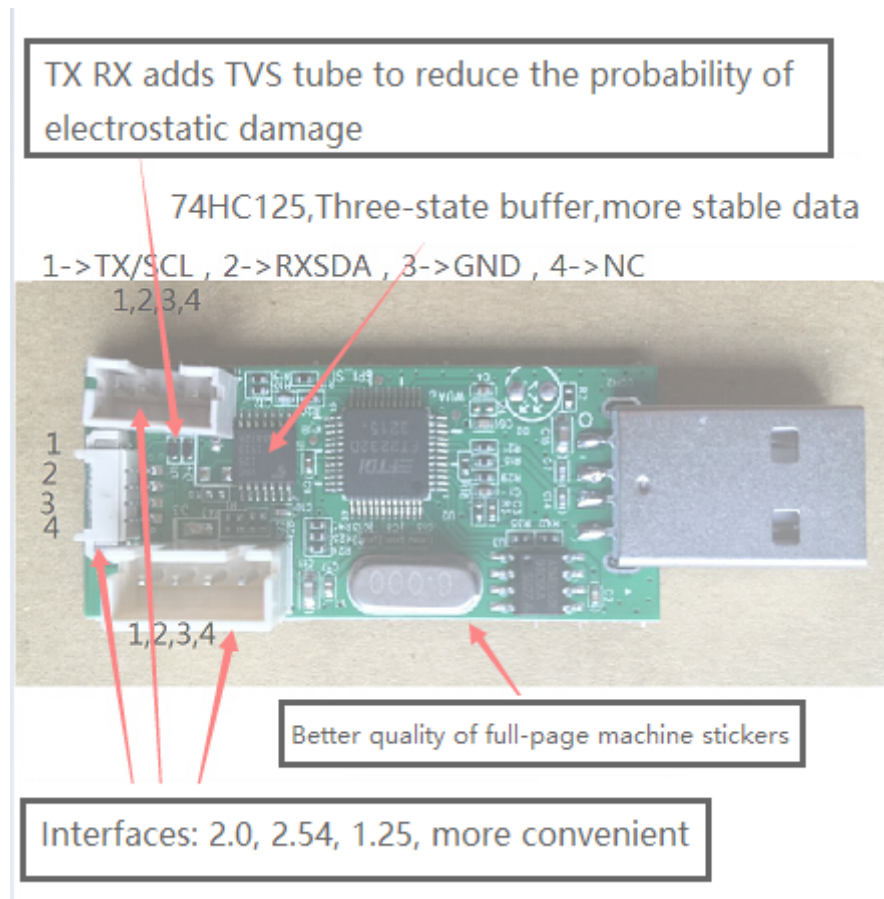


Figure 1-1

Use the USB extension cable to connect to PC, install the driver "*debug tool driver for win7*".

Open Securt CRT. Two ways to stop the serial port:

1. Enter uboot first, then debug, press Enter to **close the serial terminal** ,as shown below:

```
SigmaStar #
SigmaStar #
SigmaStar #
SigmaStar # debug
debug mode on, cmdline is disabled
SigmaStar #
```

Figure 1-2

2. Enter the system, input 11111, stop the serial port, **close the serial terminal** ,as shown below:

```

setHwcursorPhyAddr hwcursor phyaddr has been set!
[GOPI]Ha]GopUpdateGwinParam 716: GOP_id=011 not support
[GOPI]Ha]GopSetArgb1555Alpha 1201: GOPId=0x11 not support
[GOPI]Ha]GopSetArgb1555Alpha 1201: GOPId=0x11 not support
MI_ModuleDev_RegisterDev Enter

gGfxDevHdl = c10b6000.

fb0: Mstar frame buffer device
[1]+  Done                  busybox telnetd
/ #
/ #
/ #
/ #
/ # 111
-/bin/sh: 111: not found
/ # 1111
-/bin/sh: 1111: not found
/ # 11111

```

Figure 1-3

Open and confirm **chip /interface/Slave addr** in the interface , select as shown below, and then click the button, the interface shown in Figure 1-5 will appear

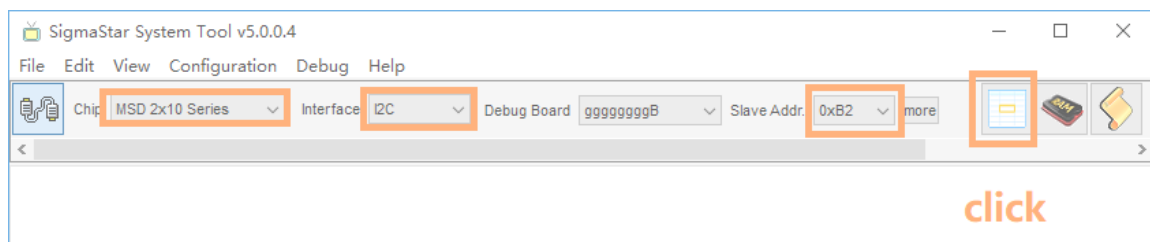


Figure 1-4

Select bank, the range is 1002-1016; select MIU, and then click Read Bank. If the data can be read normally, it means that the Sstar tool and the chip are connected normally.

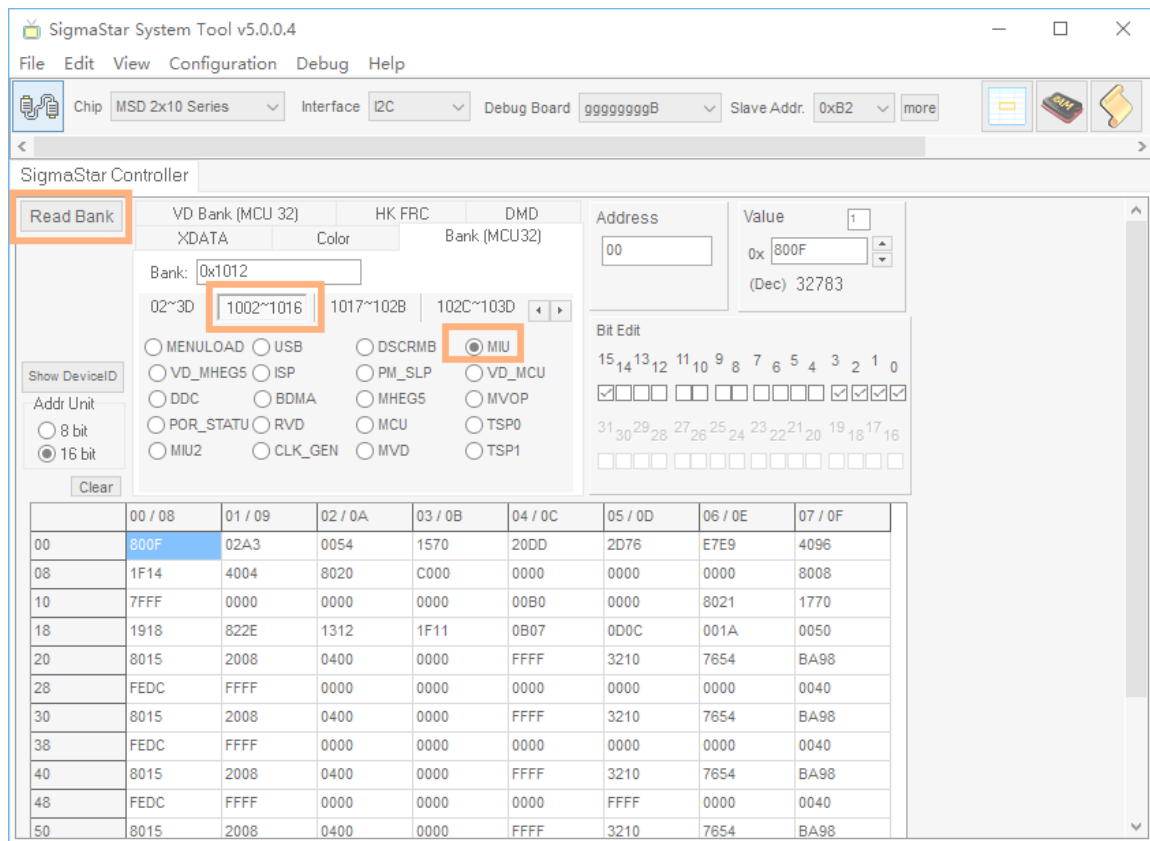


Figure 1-5

Bank1012 offset address 1B 16bit value is 0x1F11.

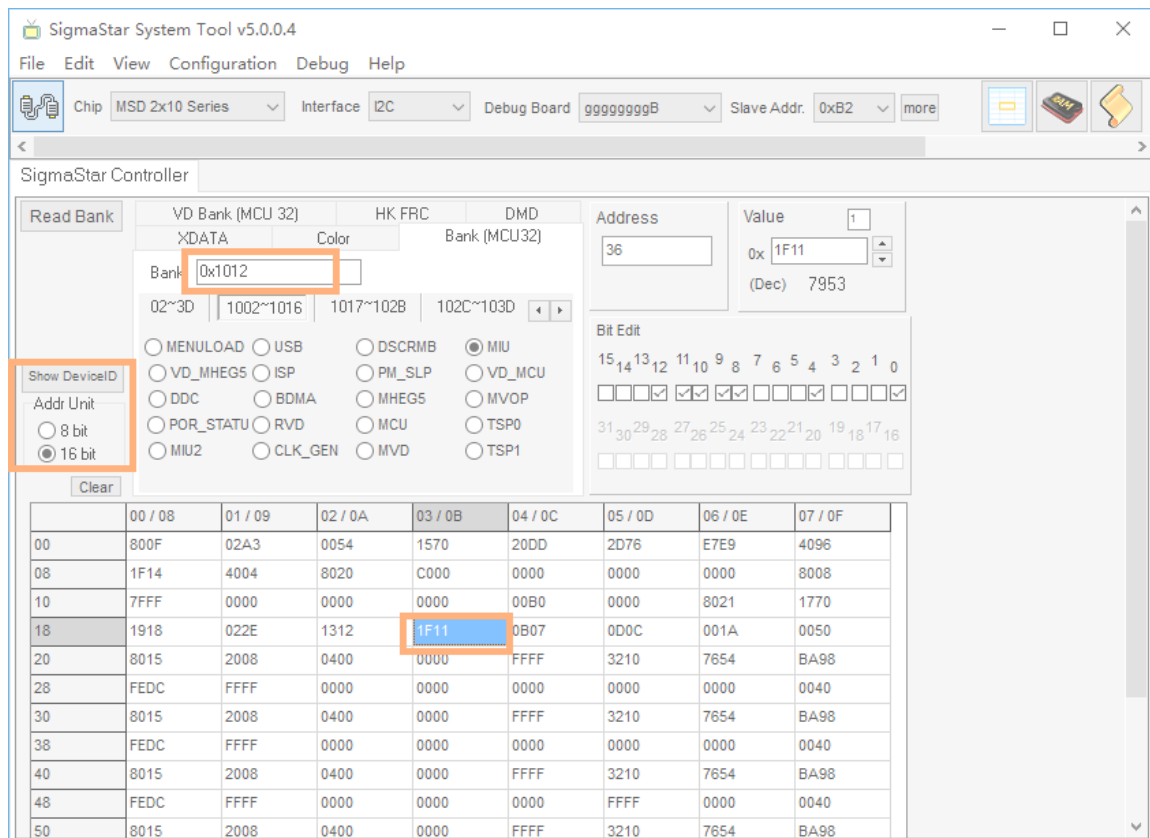


Figure 1-6

For example, Bank1012 offset address 36 8bit value is 0x1F, offset address 35 8bit value is 0x11, 8bit converted to 16bit is 36/2=1B (hexadecimal)

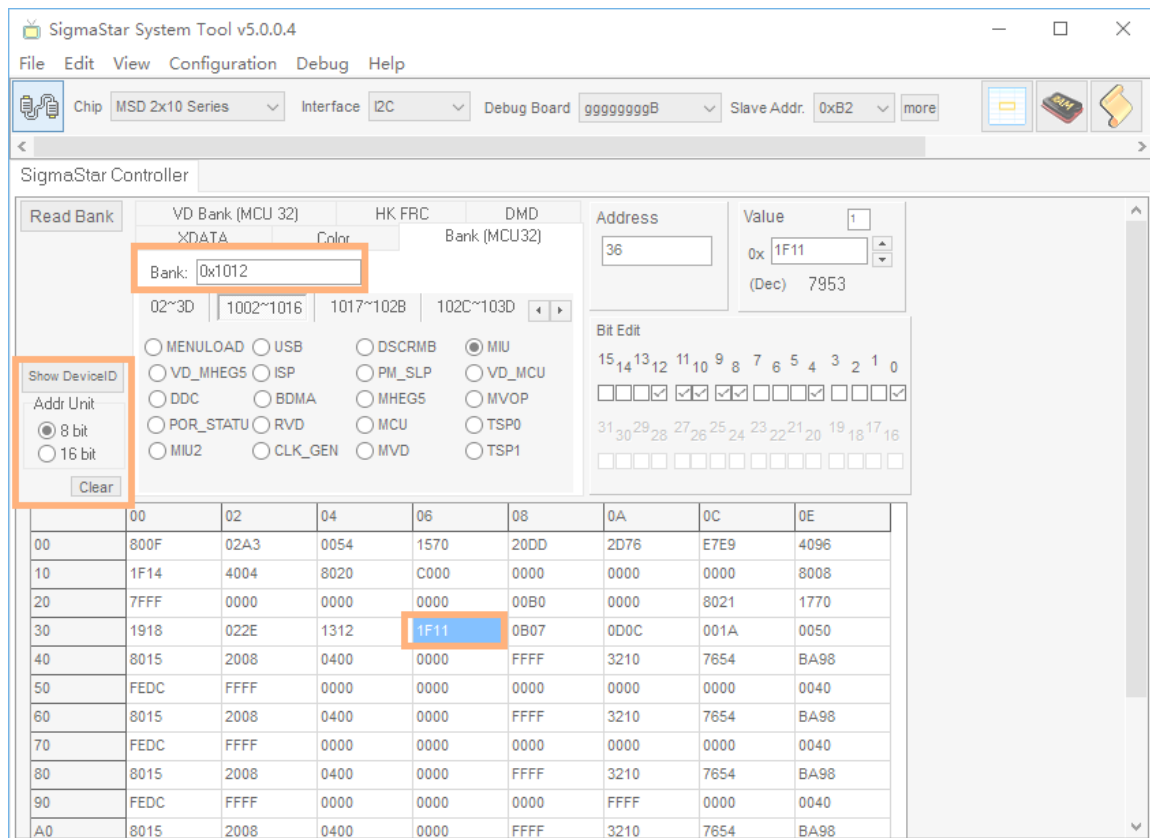


Figure 1-7

2. Sstar Flash Tool

2.1. Burning Hardware Environment Preparation

2.1.1. Debug Tool

Debug tool is a hardware serial port tool, It is dedicated to burning SigmaStar chips` uboot and *Sstar tool* register access.

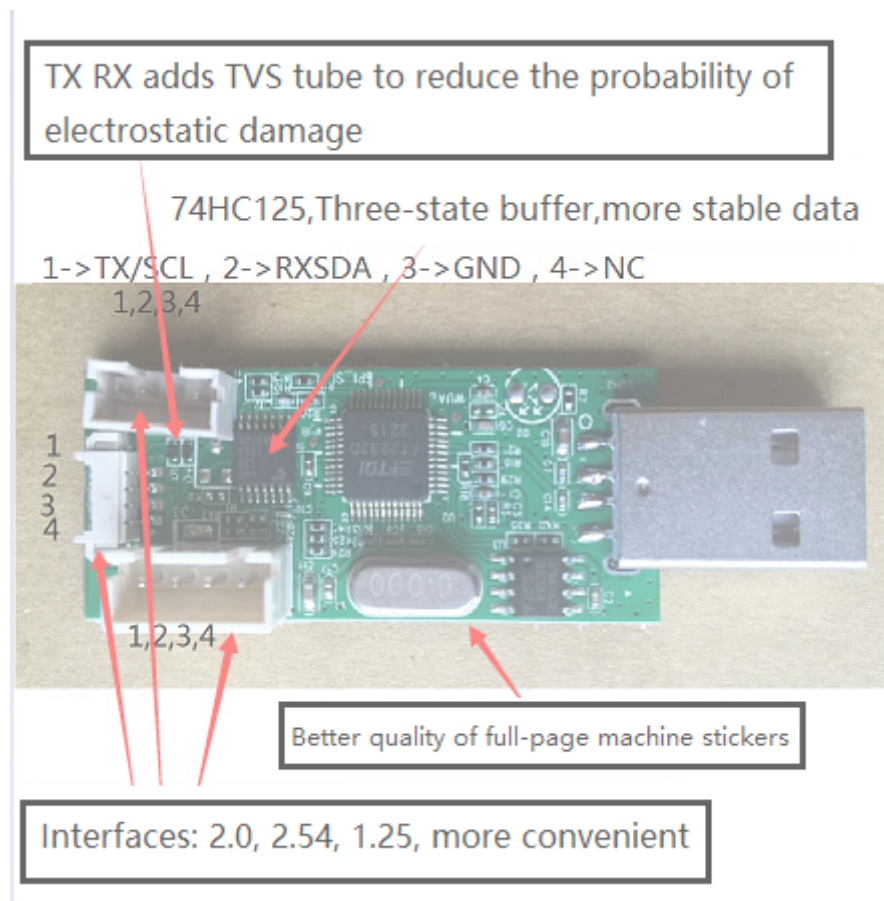


Figure 2-1

2.1.2. Hardware Block Diagram



The sequence of connecting the Sigmastar chip to the board is NC
GND RX TX

Figure 2-2

2.1.3. SPI-NOR Flash Burning

It is suitable for empty burning, or where uboot is damaged and can't be upgraded through it.

The offset burned by SPI Norflash is as follow:

Table 2-1

name	offset	path
IPL.bin	0x0000	project\image\output\i-mages\IPL.bin
IPL_CUST.bin	0x10000	project\image\output\images\IPL_CUST.bin
MXP_SF.bin	0x20000	project\image\output\images\MXP_SF.bin
u-boot.xz.img.bin	0x30000	project\image\output\images\u-boot.xz.img.bin

- Step 1: Use *debug tool* to connect the board, close the serial terminal, execute *Flash Tool*, power on the board.
- Step 2: select **SPI** tab, press **More** and then choose type **SPINOR** ;

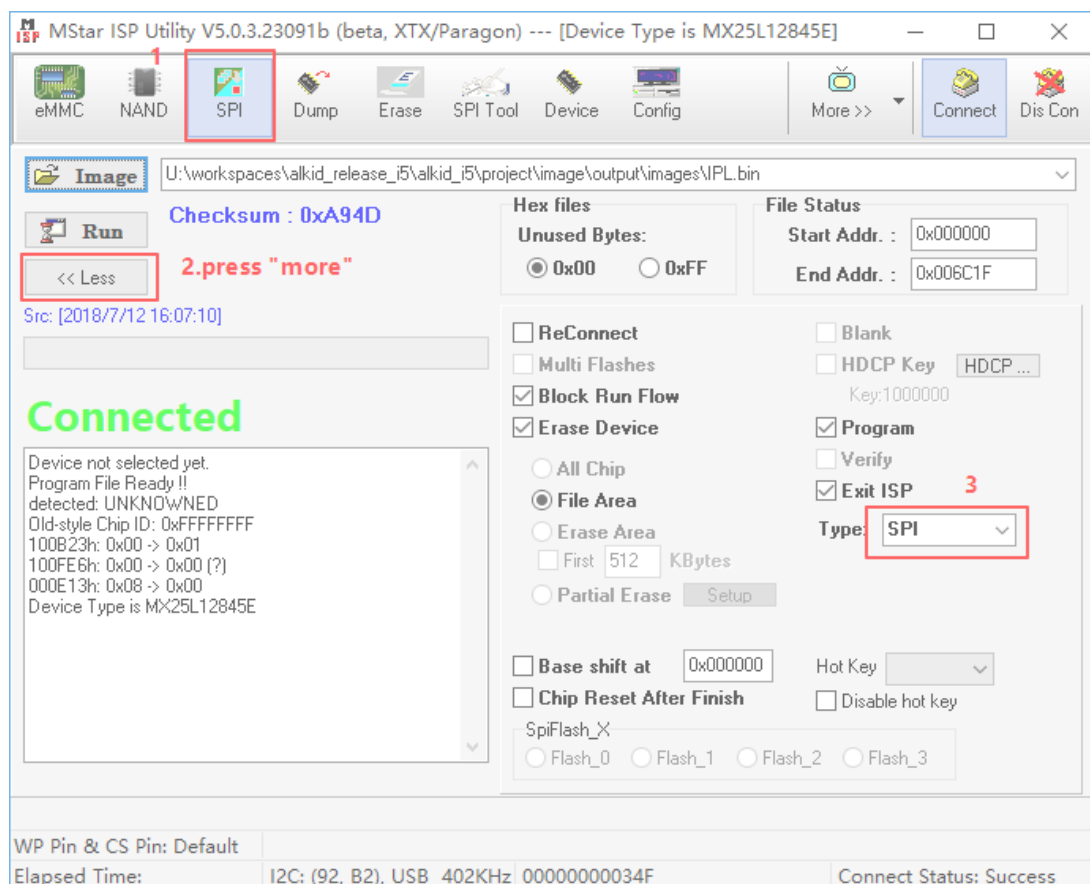


Figure 2-3

- Step 3: Load the burned file and click **Connect**

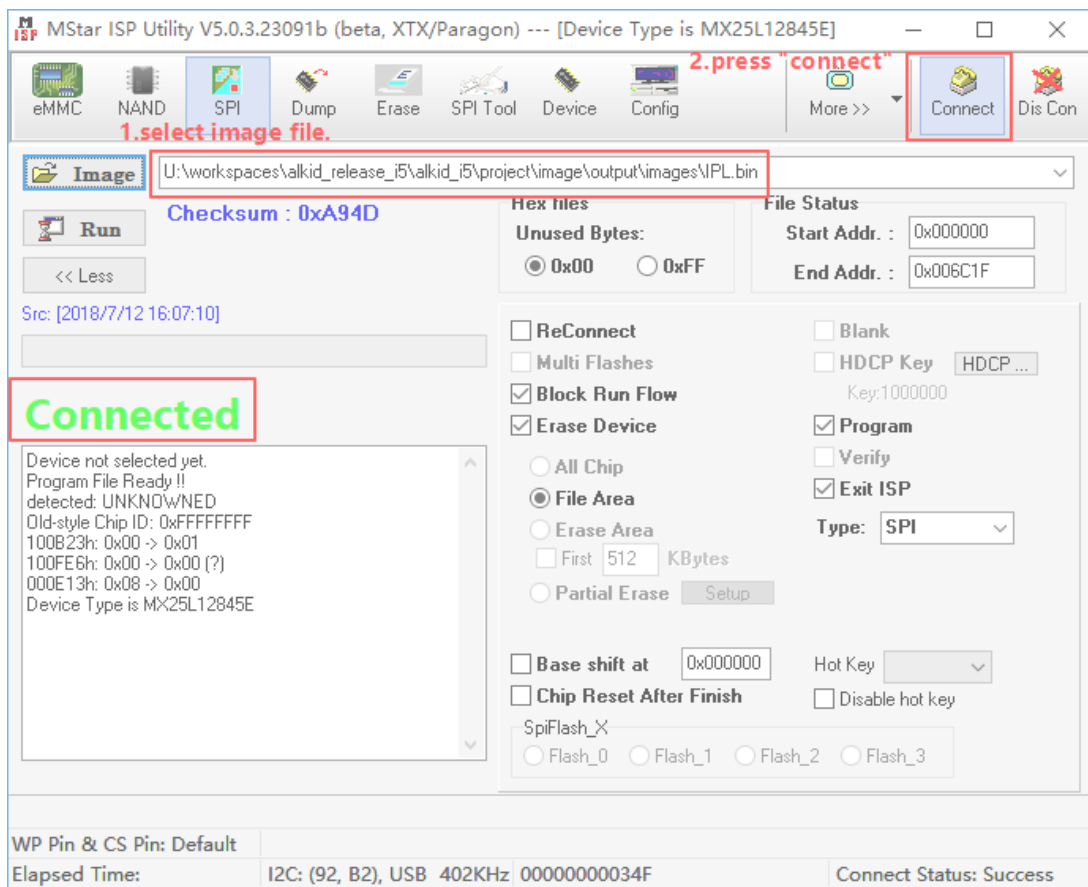


Figure 2-4

- Step 4: load **imageIPL.bin** , click **Run** ;

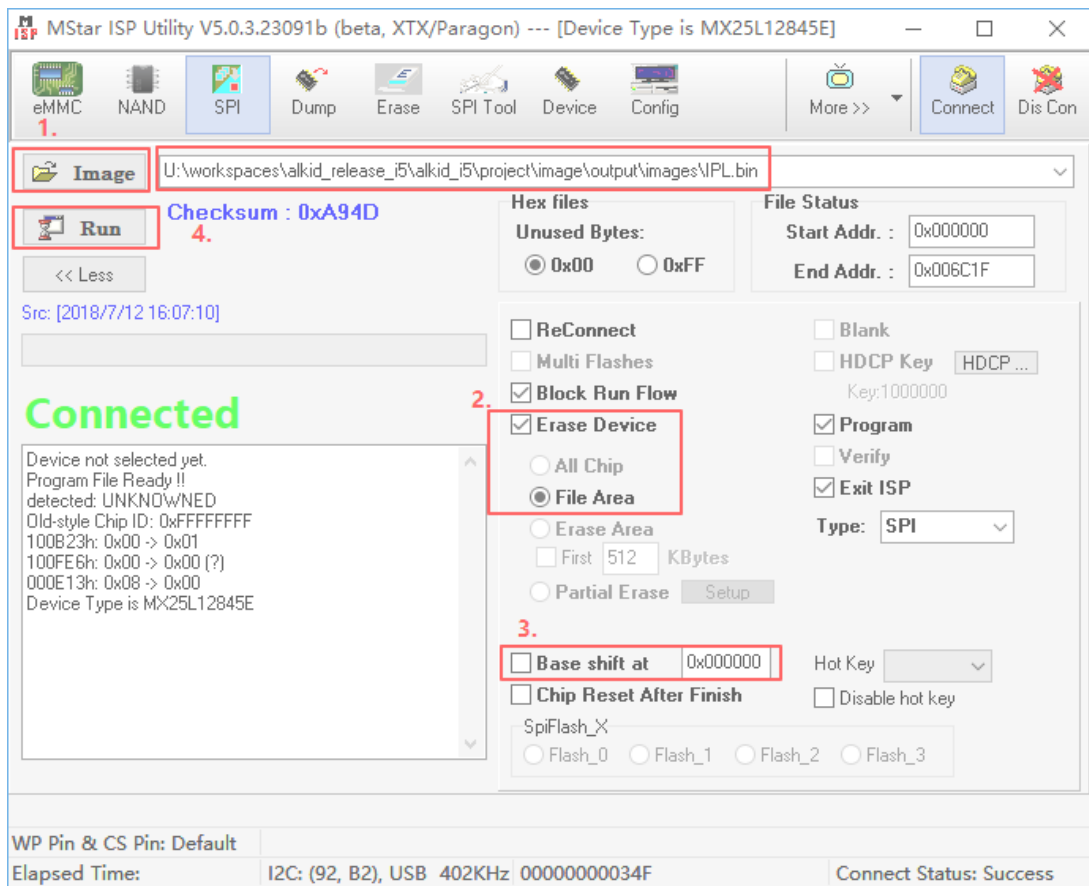


Figure 2-5

- Step 5: load image "IPL_CUST.bin", set **Base shift** at 0x10000 °

Note: It may change with the version, the address of Base shift is subject to the data in Table 2-1.

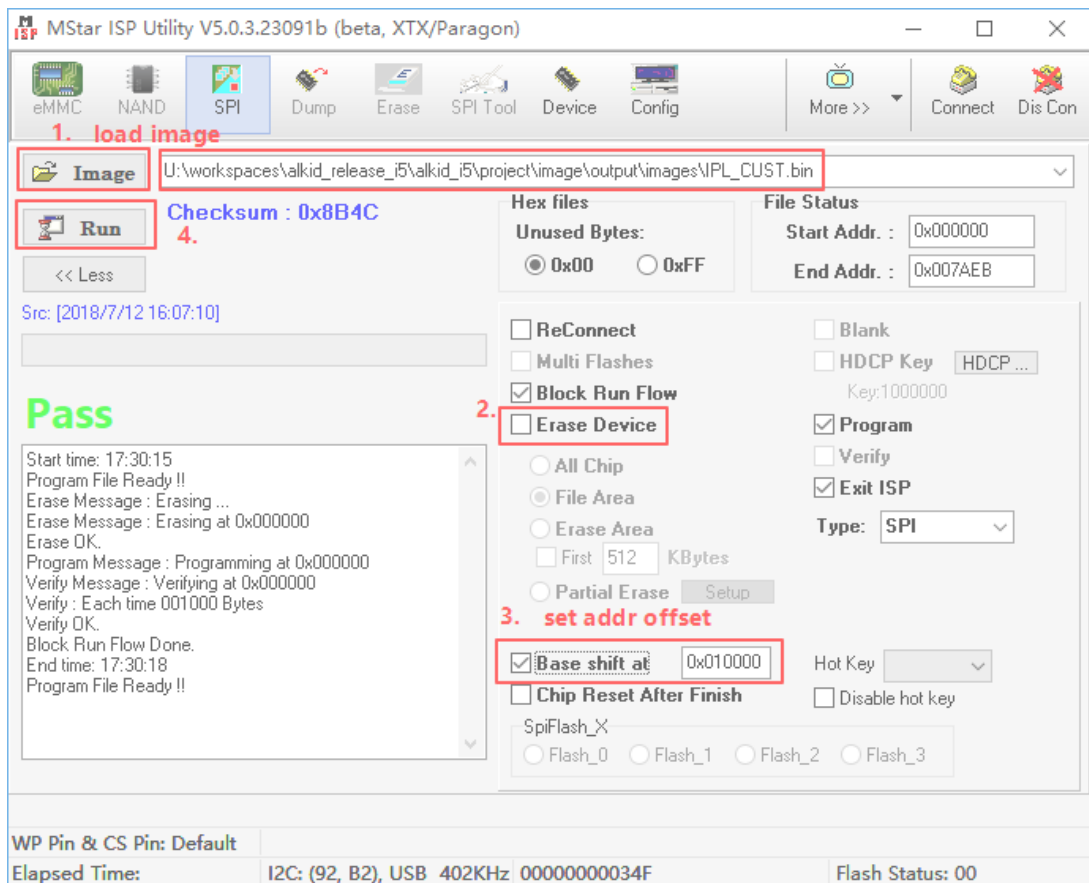


Figure 2-6

- Step 6: load image "MXP_SF.bin", set "Base shift" at 0x20000 .
- Step 7: load image "u-boot.xz.img.bin", set "Base shift" at 0x30000 .
- Step 9: restart the board

2.1.4. SPI-NAND FLASH Burning

It is suitable for empty burning, or where uboot is damaged and can't be upgraded through it.

The method of burning SPINAND is the same as SPINOR, but the address and file are slightly different.

Table 2-2

Name	Erase Device	offset	Path
GCIS.bin	All Chip	0x00000	project/l2m/image
IPL.bin	File Area	0x140000	project/l2m/image
IPL_CUST.bin	File Area	0x200000	project/l2m/image
u-boot_spinand.xz.img.bin	File Area	0x2C0000	project/l2m/image

- Step 1: Use *debug tool* to connect the board, close the serial terminal, power on the board.
- Step 2: Open *Flash Tool*, as shown below:

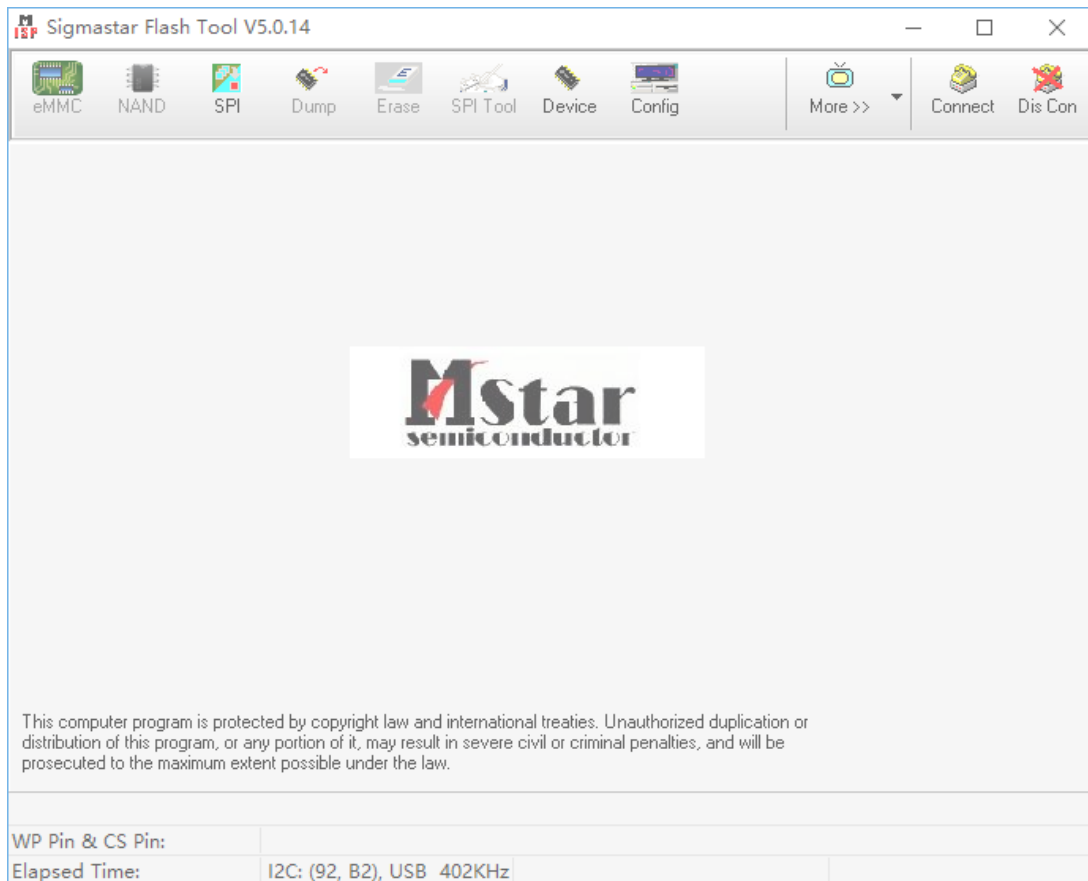


Figure 2-7

- Step 3: select the SPI button, as shown below:

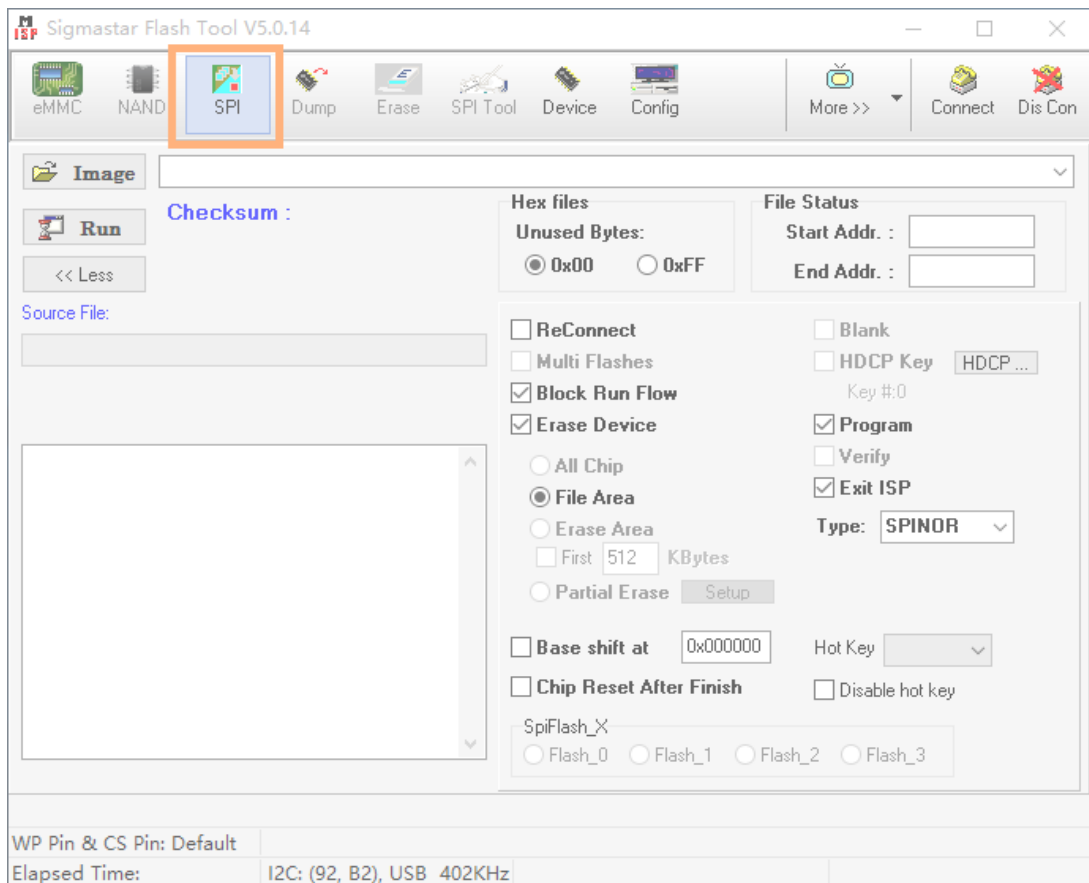


Figure 2-8

- Step 4: select type SPINAND, click *Connect*, and then click OK.

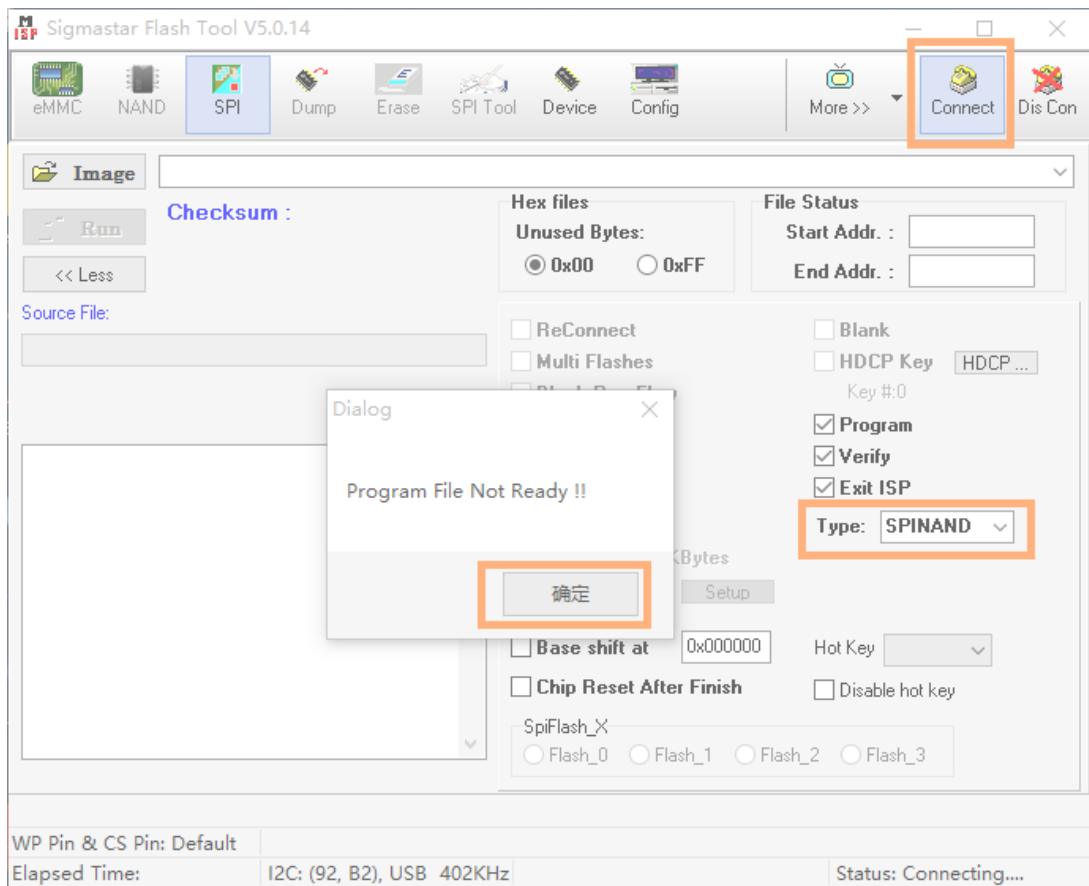


Figure 2-9

- Step 5: When the corresponding Flash model is displayed, the connection is successful. as shown below:

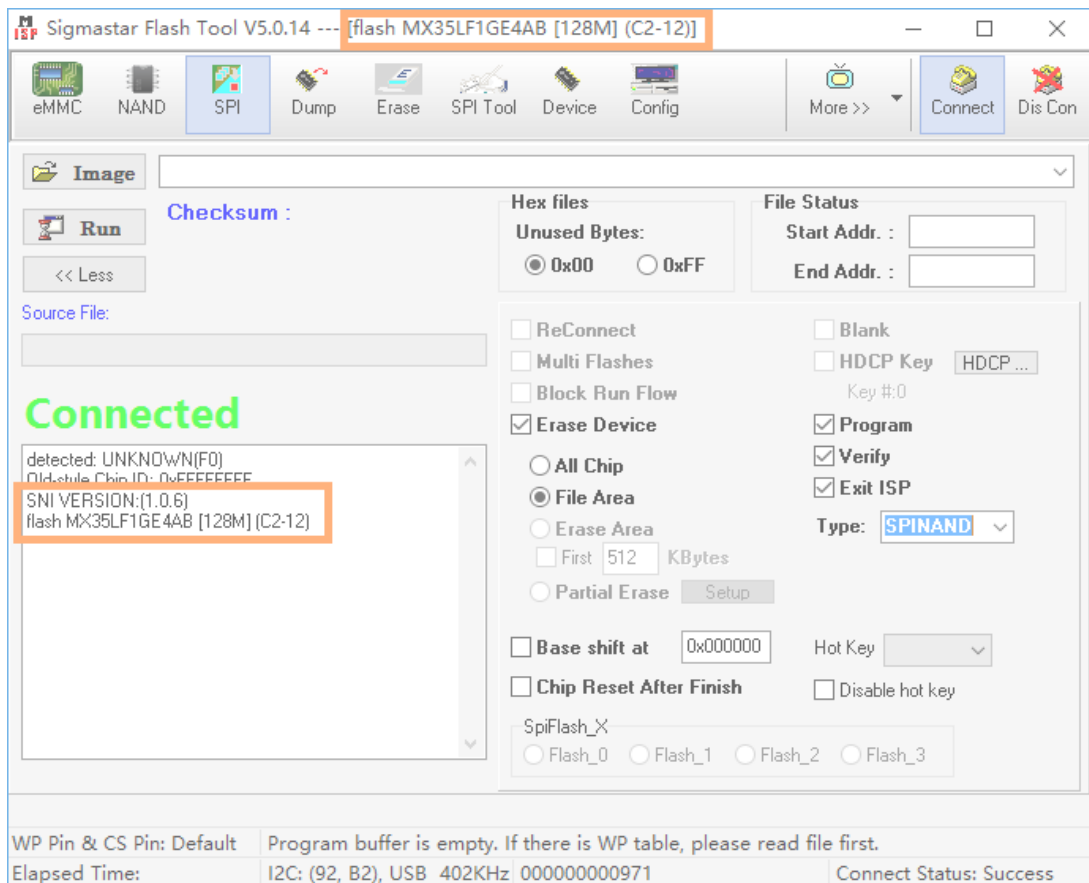


Figure 2-10

- Step 6: load image "GCIS.bin", click "Run" ;

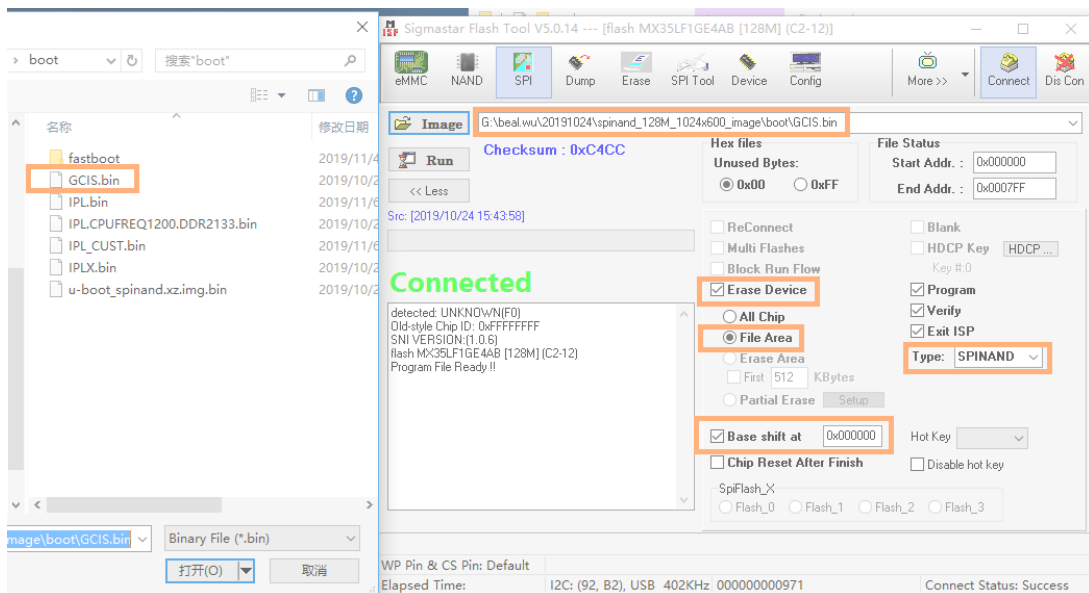


Figure 2-11

- Step 7: load image "IPL.bin", set "Base shift" at 0x140000 .

Note: It may change with the version, the address of Base shift is subject to the data in Table 2-2.

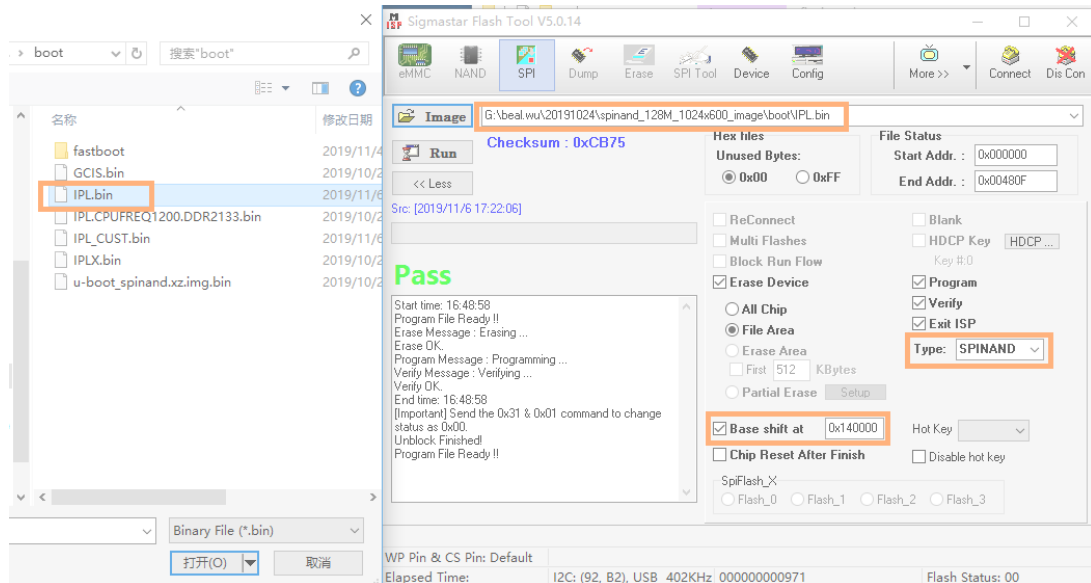


Figure 2-12

- Step 6: load image “IPL_CUST.bin”, set “Base shift”at 0x200000 °

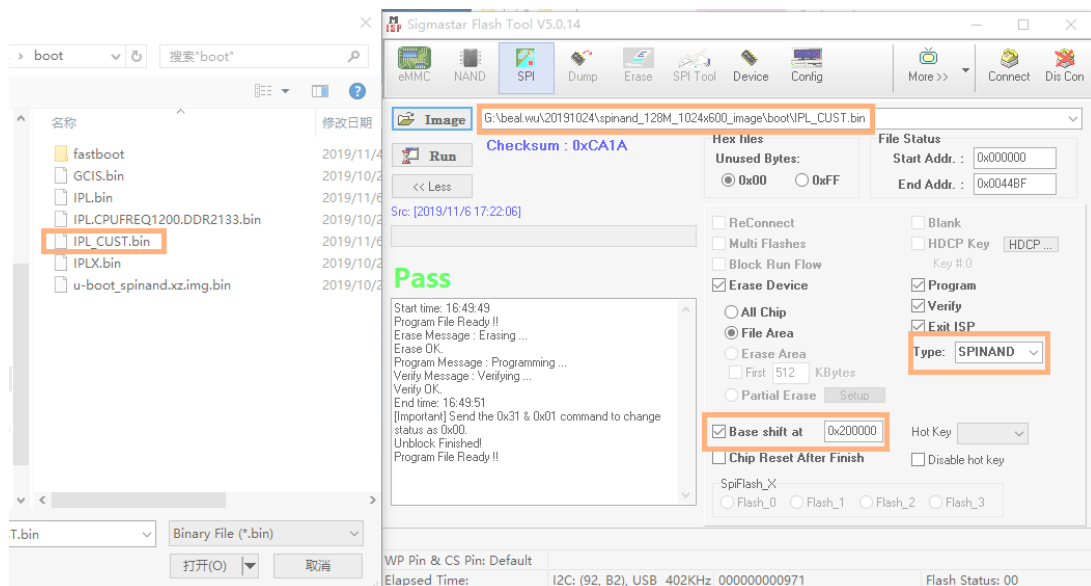


Figure 2-13

- Step 7: load image “u-boot_spinand.xz.img.bin” , set “Base shift”at 0x2c0000 °

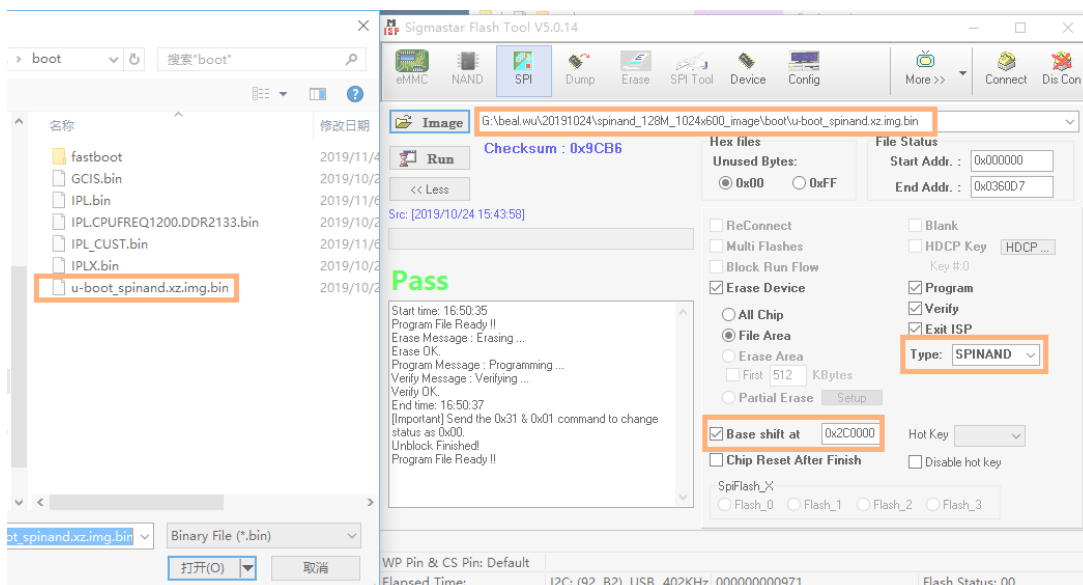


Figure 2-14

- Step 9: restart the board

2.1.5. Image Burning Method

After using the Flash tool to burn uboot, there are three ways to burn the image file: USB, SD card, and network. Different chips support different burning methods. Here we focus on the most common burning method for software development — network Burning.

We need to prepare PC, switch, development board, network topology for network burning.

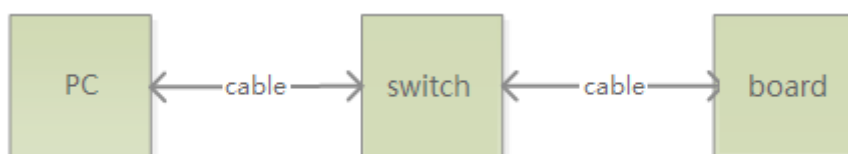


Figure 2-15

- step1 : ensure that the network connection is normal, set the development board IP and server IP, and then save the environment variables.

```

In:    serial
Out:   serial
Err:   serial
Net:   MAC Address 00:30:1B:BA:02:DB
Auto-Negotiation
Link Status Speed:100 Full-duplex:1 Network connection is normal
sstar_emac
Warning: sstar_emac using MAC address from net device

SigmaStar # setenv ipaddr 172.19.24.5 Development Board IP
SigmaStar # setenv serverip 172.19.24.145 Server IP
SigmaStar # saveenv
Saving Environment to NAND...
SPINAND: MDrv_SPINAND_GetPartOffset: UBOOT_PBA==0 and no PNI: 0 0 0
SPINAND: MDrv_SPINAND_GetPartOffset: use offset 1E0000
Erasing NAND...
Erasing at 0x1e0000 -- 100% complete.
Writing to NAND... OK IP saved successfully
SigmaStar #

```

Figure 2-16

- step2 : use the Image burning path specified by TFTP to ensure that the server IP is configured correctly, as shown below, the board Serverip is 172.19.24.149 .

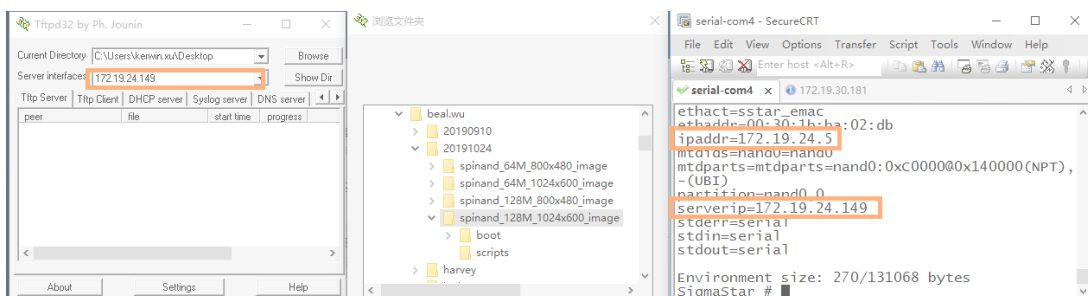


Figure 2-17

- step3 : input "estar" command to burn.

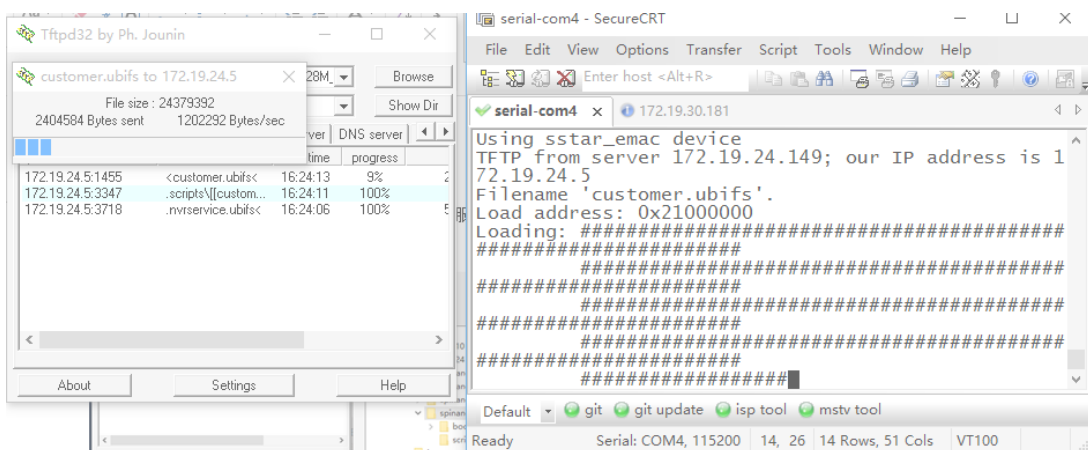


Figure 2-18

2.1.6. How to clear Flash

Sometimes there are programs in the flash and need to be erased with the Flash tool software. As shown below, select the correct type of Flash, then select "All Chip", specify a **GCIS.bin** file, and click "Run".

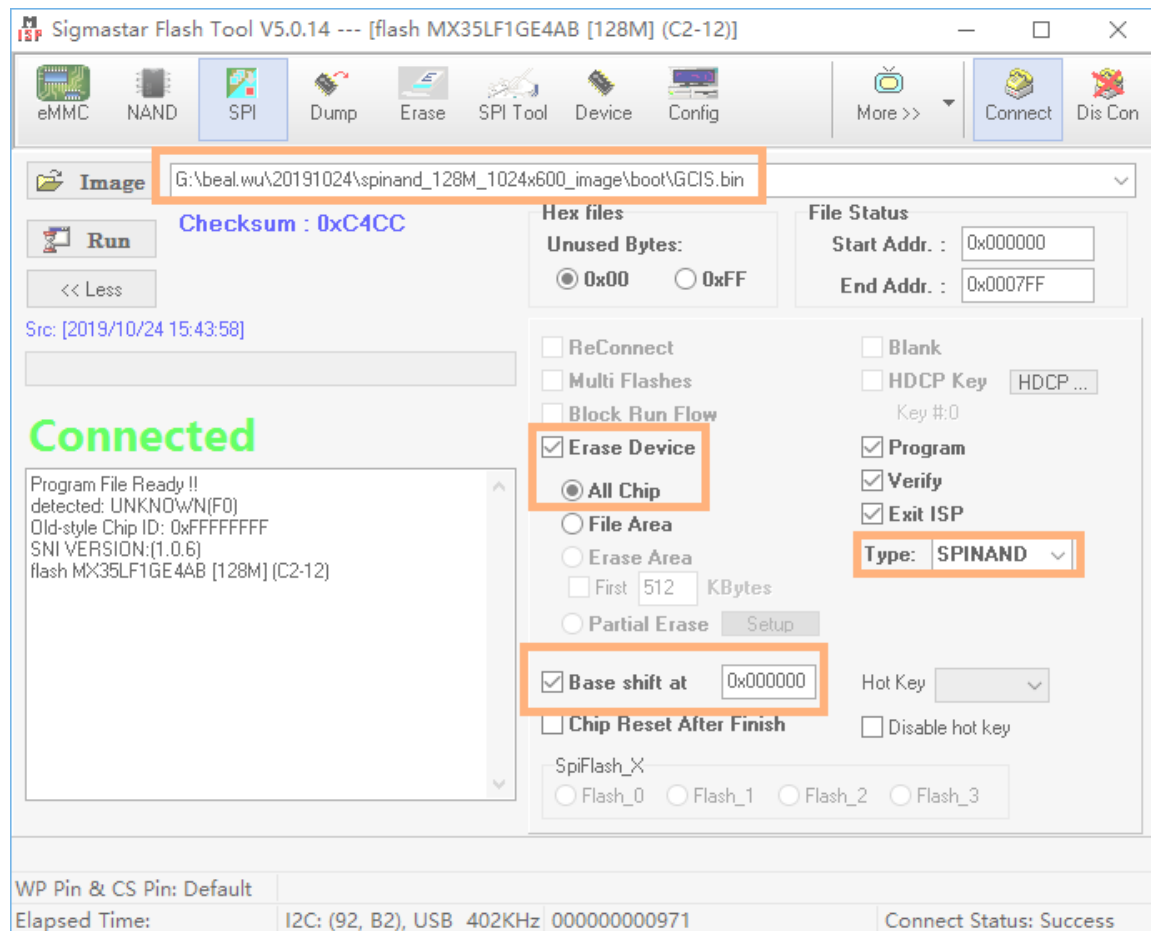


Figure 2-19

2.2. Sstar Flash Tool Burning Error

2.2.1. Hardware Connection Error

Use Debug tool connect with chip, please pay attention to the wiring sequence of the serial port.

SigmaStar's chip debug serial port Pin is **PM_UART_RX/PM_UART_TX**, the interface sequence is NC GND RX TX, pay attention to the name of the Debug Tool PCBA silkscreen, the connection method is RX to RX, TX to TX, GND to GND.

2.2.2. Serial Port Tool Selection Error

Empty burning doesn't use debug tool, but uses ordinary serial port tools, as shown below.

SigmaStar's chip debugging serial port has I2C function inside, which is used when programming access registers. But common serial port printing tool doesn't have it, so you need to use the debug tool mentioned in Figure 1-1.



Figure 2-20

2.2.3. Flash Type Selection Error

SPInand doesn't select the corresponding Flash type, such as SPINAND, but SPINOR is selected. As shown below, the wrong Flash type is selected, the chip shows connected, but the connection is not successful in fact.

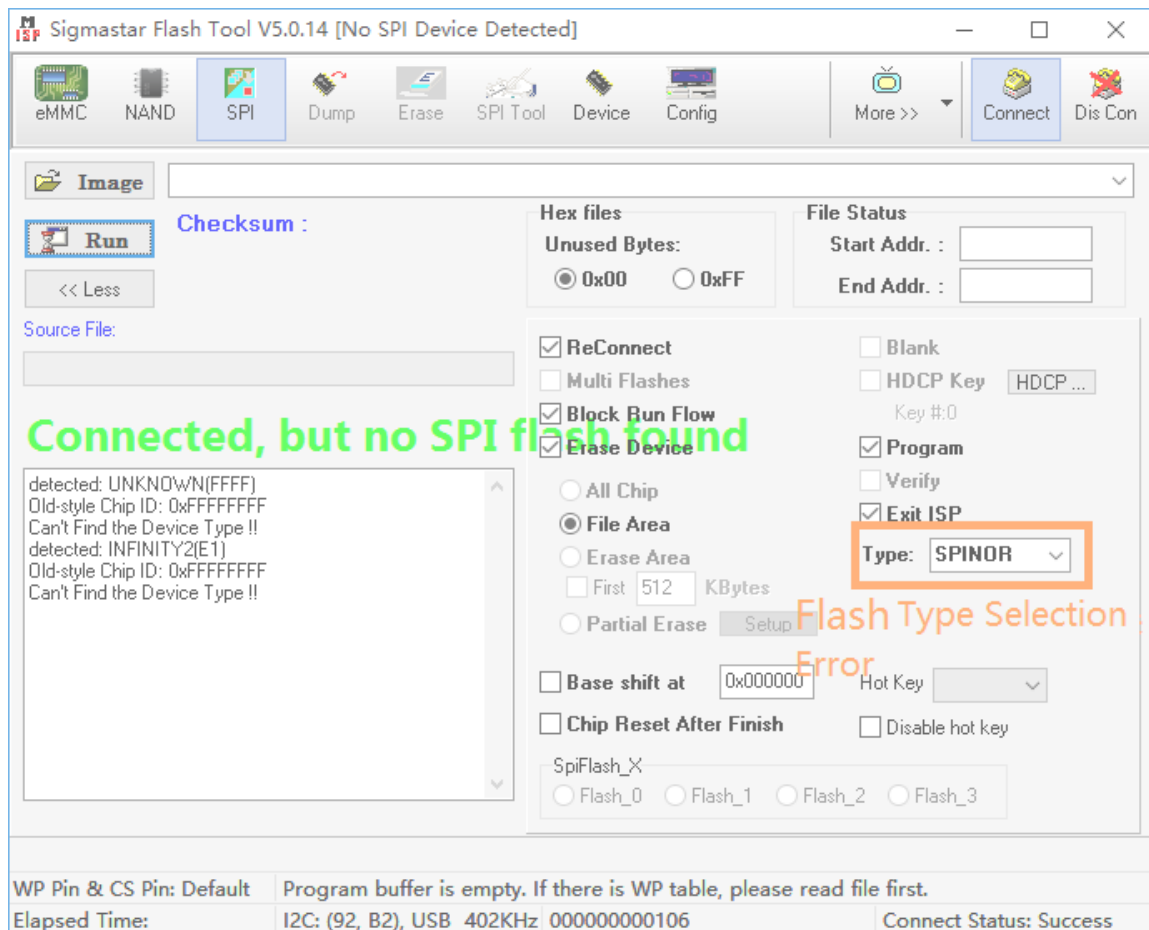


Figure 2-21

2.2.4. Flash Tool File Lack

The Flash type is selected correctly, but the SPINAND model is still not found. It may be that the file `SPINANDINFO.sn1` is missing, which contains information such as SPINAND manufacturer and ID.

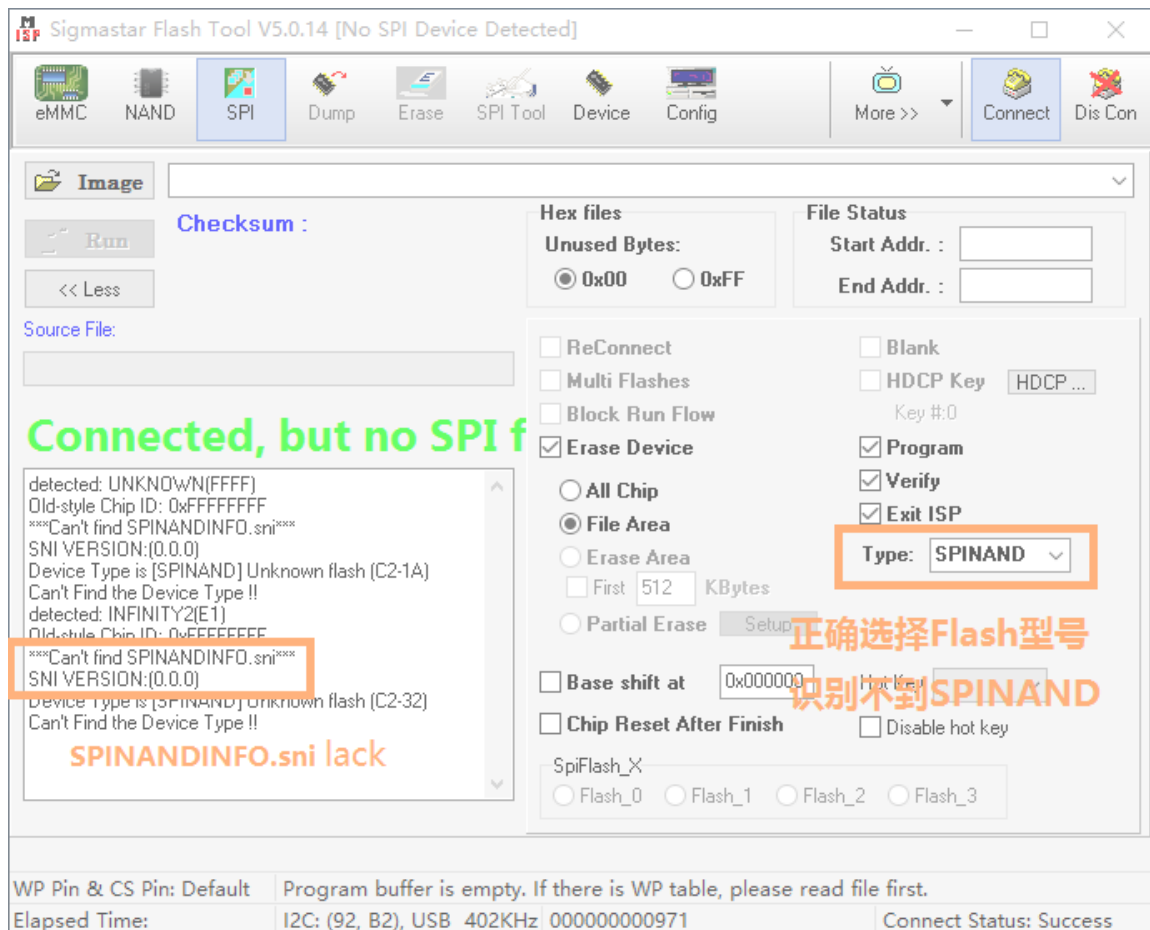


Figure 2-22

2.2.5. Serial Terminal Open

Open the Flash tool and the serial terminal is turned on, causing the Flash tool to fail to recognize Flash.

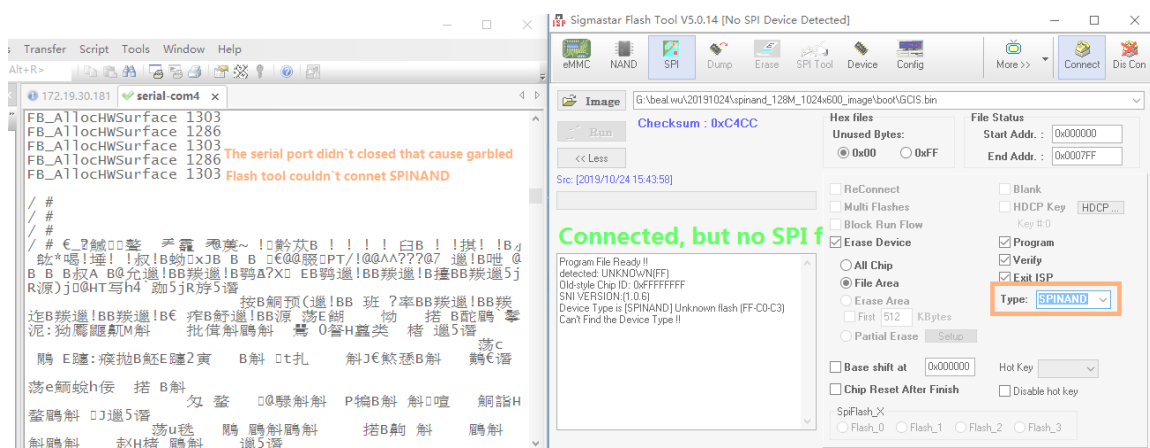


Figure 2-23